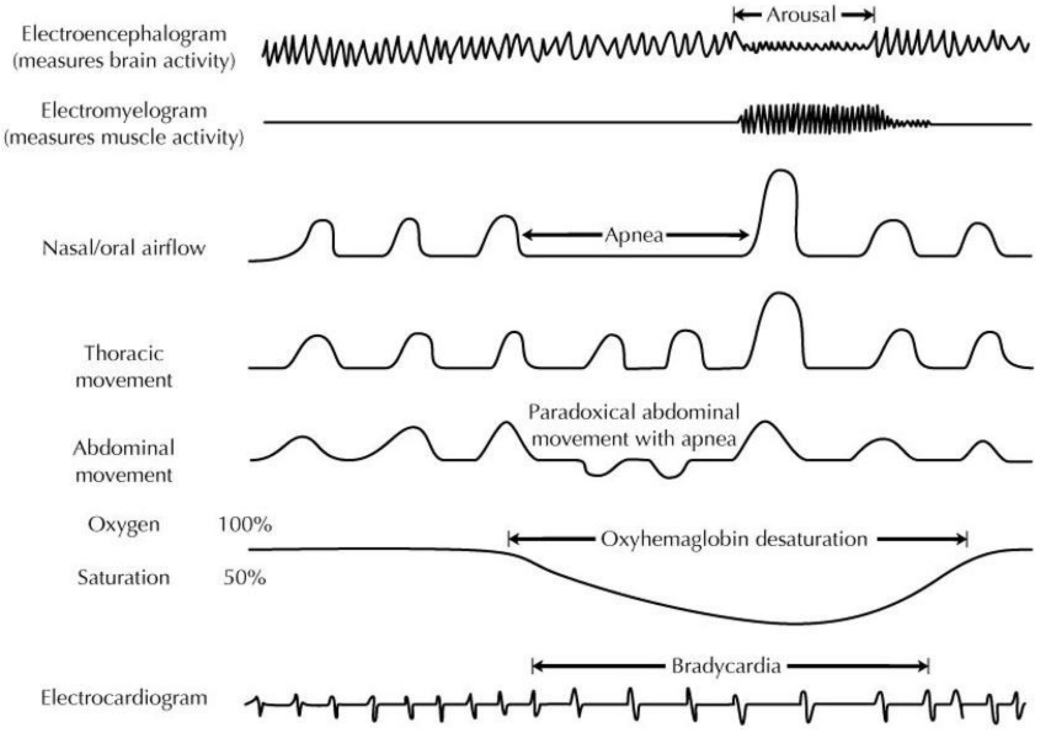
OSA: The physiological problem

- Characterized by **recurrent** episodes of upper airway partial or complete obstruction
- Caused by **upper airway collapse** during sleep
- Characterized by **frequent awakenings** caused by apnea and/or hypoapnea

Peng et al. 2021

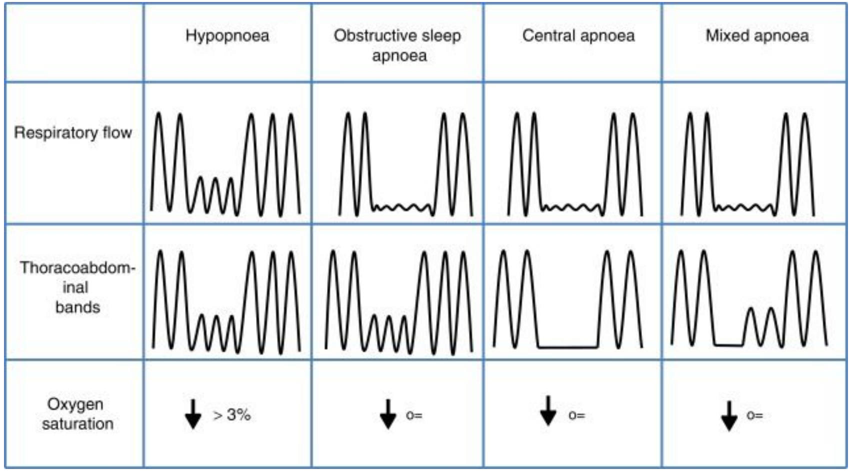
The clinical problem: Polysomnography to characterize OSA



Warvedaker NV et al. Best Practice of Medicine. Sept. 1999

- Poly-somnus: Multi channel measurement of:
  - brain waves
  - oxygen levels
  - respiratory effort
  - nasal airflow
  - heart rate
  - snoring sound
  - limb movement
  - …
- OSA characterized by presence of apnea and hypoapnea

The clinical problem: Polysomnography to characterize OSA



Apnea-Hypopnea index (AHI) = Number of apnea + hypoapnea events per hour of sleep

AHI based classification:

- < 5 Normal
- > 5 and < 15 Mild
- > 15 and < 30 Moderate
- > 30 Severe

Hypoapnea: ↓ flow ≥ 30% from baseline for at least 10 seconds with 3% O2 desaturation OR arousal

Apnea: ↓ flow ≥ 90% from baseline for at least 10 seconds

[Ramos et al 2016](#)

The analytical problem

- Patient plugged onto the machine for a duration of **<u>8 hours</u>**
- Multiple channels (32!) are measured per patient → no clarity on the importance of measuring one or more
- Technician manually scan through the 8 hours of multi-channel records before assigning a final AHI score

<u>Questions</u>:

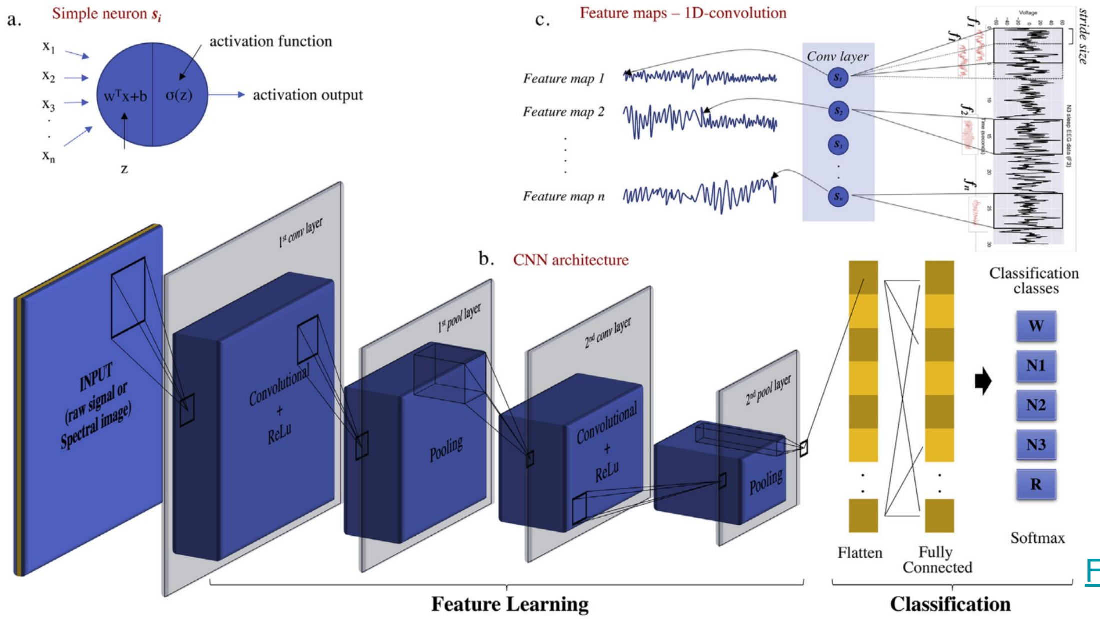
- What are the most “informative” channels for classifying the severity of OSA (AHI score)?
- Can the manual scoring step be made more efficient?
- Can the “technological solution” be universally deployed?
- Are clinical covariates a good predictor of severity?

Wishlist for an ‘ideal’ OSA test

- **Frugal**: Only a subset - the most informative channels are monitored
- A **‘summarized’** view - at what time points is apnea/hypoapnea triggered; what is the correlation structure with other channels
- Easy to **run/deploy** on a typical lab computer
- **Interpretable**

Potential gaps that AI/ML can fill in OSA detection

- Identify channels of importance by scanning through data over thousands of retrospective scans
- Identify “change points” where apnea/hypoapnea is triggered



[Fiorollo et al. 2019]

What has been tried?

Research Article

A Clinical Prediction Formula for Apnea-Hypopnea Index

Mustafa Sahin,[1] Cem Bilgen,[2] M. Sezai Tasbakan,[3] Rasit Midilli,[2] and Ozen K. Basoglu[3]

[1] *Department of Otorhinolaryngology, Diskapi Yildirim Beyazit Research and Training Hospital, Irfan Bastug Street, Dıskapi, 06110 Ankara, Turkey*
[2] *Department of Otorhinolaryngology, Ege University School of Medicine, İzmir, Turkey*
[3] *Department of Chest Diseases, Ege University School of Medicine, İzmir, Turkey*

Correspondence should be addressed to Mustafa Sahin; iskebaha@gmail.com

nature communications

Article https://doi.org/10.1038/s41467-023-40604-3

Deep learning for obstructive sleep apnea diagnosis based on single channel oximetry

Received: 1 February 2023

Accepted: 3 August 2023

Jeremy Levy[1,2], Daniel Álvarez[3,4,5], Félix Del Campo[3,4,5] & Joachim A. Behar[2]

Uses demographic, anthropometric and physical data for prediction of AHI

Used just the oximetry channel of PSG to predict the AHI score

One approach might not fit all…

nature communications

Article

https://doi.org/10.1038/s41467-023-40604-3

Deep learning for obstructive sleep apnea diagnosis based on single channel oximetry

Received: 1 February 2023

Accepted: 3 August 2023

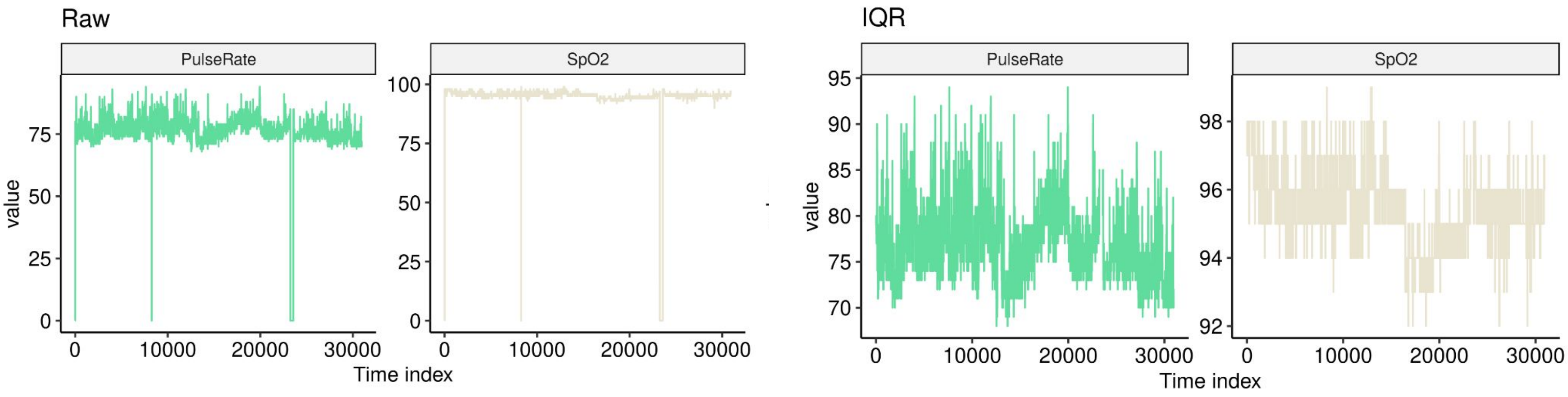
Published online: 12 August 2023

Jeremy Levy[1,2], Daniel Álvarez[3,4,5], Félix Del Campo[3,4,5] & Joachim A. Behar[2]

ticipants (Table S3). Thus, while OxiNet performed well on white and Chinese American participants, it performed poorer on Hispanic, Black, and African American participants. These results are consistent with the melanin and typology angle levels used to characterize skin pigmentation in different ethnic groups[18]. It emphasizes the lack of inclusion or low prevalence of such minority groups in datasets traditionally used to train DL algorithms which inevitably leads to embedded biases and poorer diagnostic performance for these groups. Recent research[19,20] has shown that pulse oximeter performance discrepancies have been shown to be affected by patients of different races and ethnicities, leading to poorer clinical management. This may explain the poorer performance of the model on Black and African American participants. The drop in performance in UHV was

Workflow for an ‘ideal’ **OSA test**

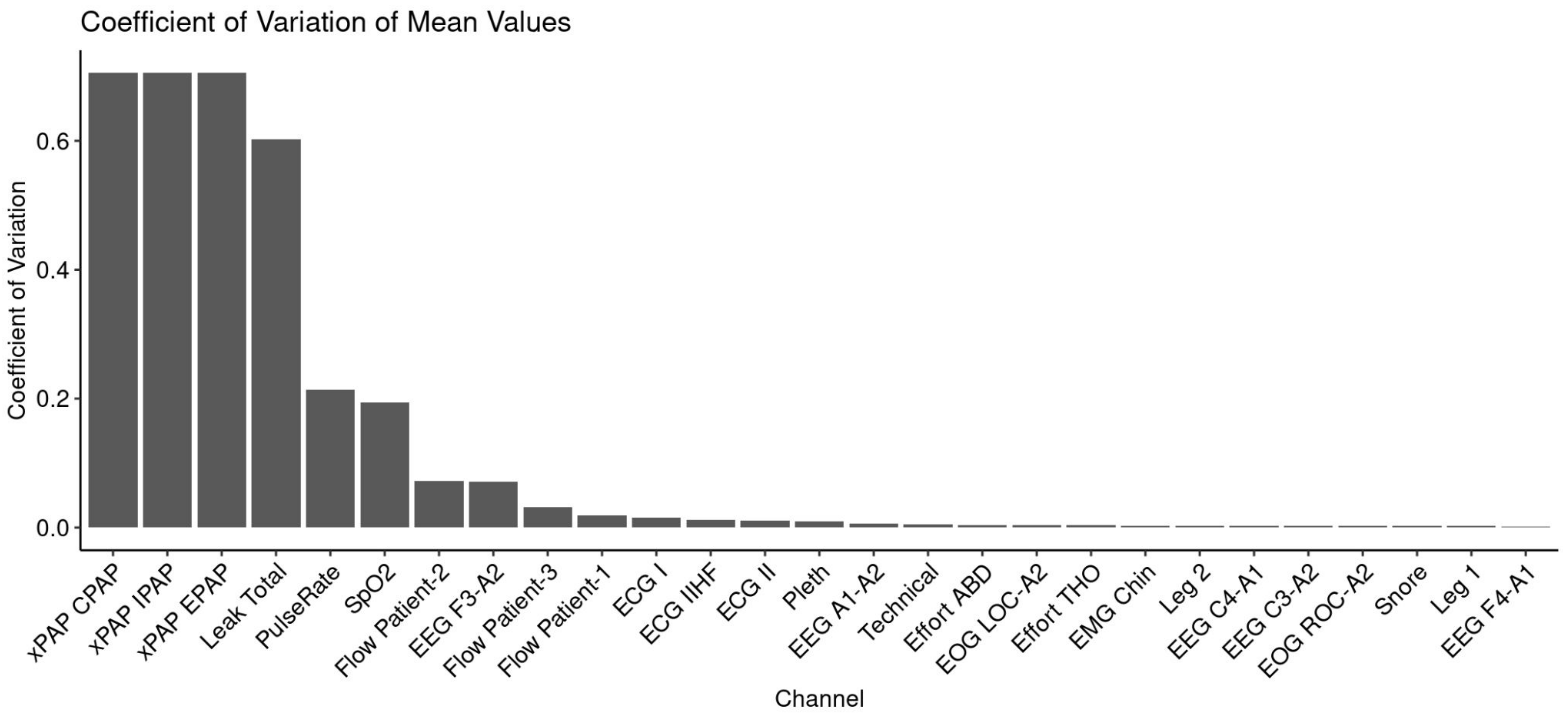
Step 1: Clearing outliers



Data: Dr. Saneev Sinha, AIIMS

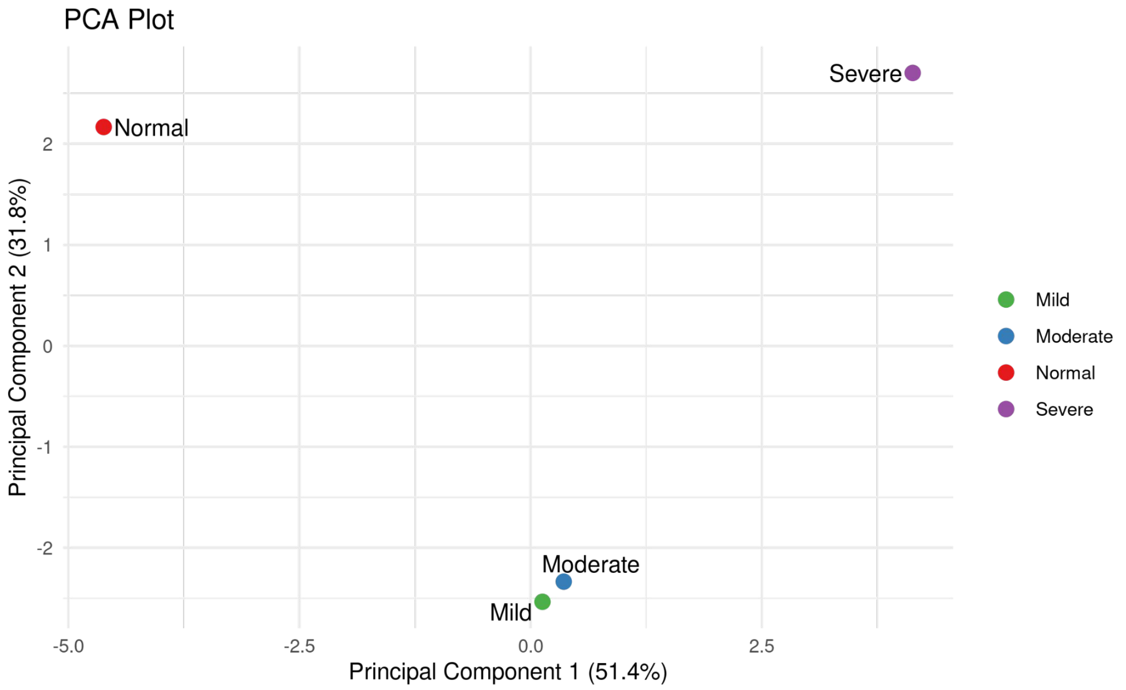
Data: Dr. Saneev Sinha, AIIMS

Step 2: Identifying features that matter

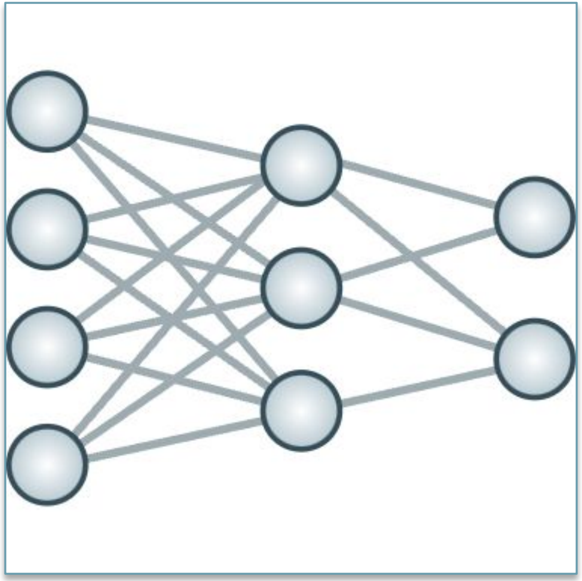


Data: Dr. Saneev Sinha, AIIMS

Step 3: Learning the differences across conditions

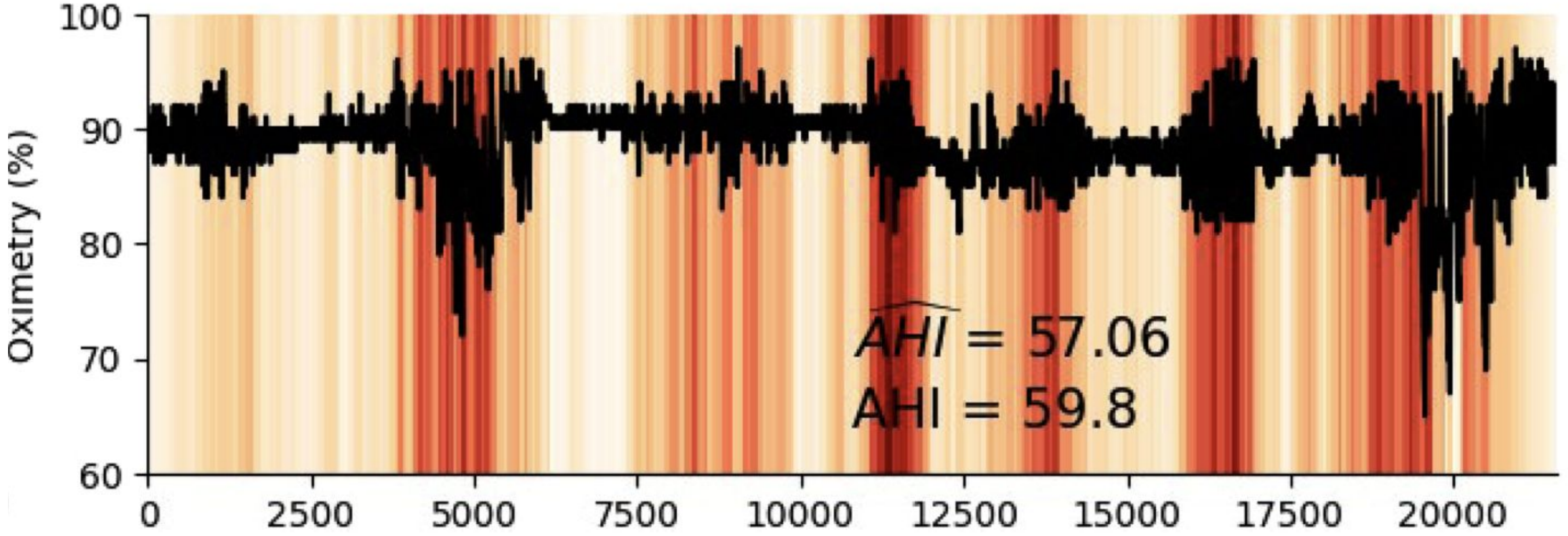


vs



+ Clinical covariates

Step 4: Identifying areas of focus



Summary

- Polysomnography tests is a multi-channel test for diagnosis the severity of obstructive sleep apnea
- AI/ML methods can make a real impact:
  - reduce “channel load” (for the patient)
  - reduce “cognitive load” (for the technician)
  - reduce diagnosis time
- An ideal solution would be:
  - interpretable
  - easily deployable
  - frugal